

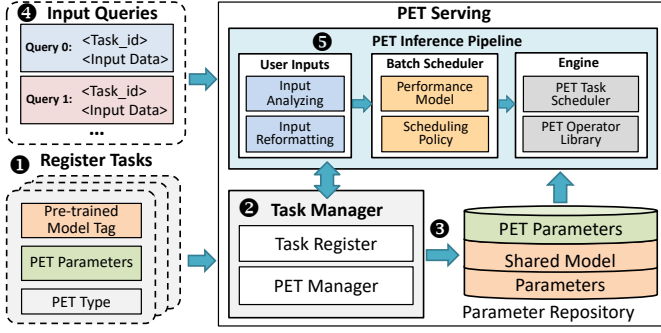


Fig. 12: PetS system overview: (1) Tasks register; (2) Task manager (3) Task schedule; (4) Task serving. (Image is taken from PetS [235])

b) Efficient kernel design: Punica addresses the first challenge by using existing matrix multiplication for the backbone computation and introducing a new CUDA kernel, Segmented Gather Matrix-Vector Multiplication (SGMV), for adding the PEFT add-ons to the backbone computation in a batched manner. This kernel parallelizes the feature-weight multiplication for different requests in the batch and groups requests corresponding to the same PEFT model to increase operational intensity and use GPU Tensor Cores for acceleration.

The second challenge is beyond the computational cost, designing an efficient system architecture that can effectively serve multi-tenant PEFT model workloads on the smallest set of GPUs possible while occupying the least amount of GPU resources is another significant challenge. Punica addresses this by scheduling user requests to active GPUs that already serve or train PEFT models, thereby improving GPU utilization. For older requests, Punica periodically migrates them to consolidate workloads, thus freeing up GPU resources for new requests.

c) Multi-Tenant PEFT design: Designing an efficient system for the multi-tenant PEFT model serving in the Punica framework focuses on addressing several key challenges to maximize hardware utilization and minimize resource consumption. The system aims to consolidate multi-tenant LoRA serving workloads onto the smallest set of GPUs possible. This consolidation is achieved through strategic scheduling of user requests to active GPUs that are already serving or training LoRA models, thereby improving GPU utilization. For older requests, Punica periodically migrates them to consolidate workloads further, thus freeing up GPU resources for new requests. It incorporates on-demand loading of LoRA model weights, which introduces only millisecond-level latency. This feature provides Punica with the flexibility to dynamically consolidate user requests to a small set of GPUs, without being constrained by the specific LoRA models already running on those GPUs. Besides that, Punica identifies that the decode stage is a predominant factor in the cost of model serving, Punica's design primarily focuses on optimizing decode stage performance. Other aspects of model serving leverage straightforward techniques, such as on-demand loading of LoRA model weights, to efficiently manage resource utilization.

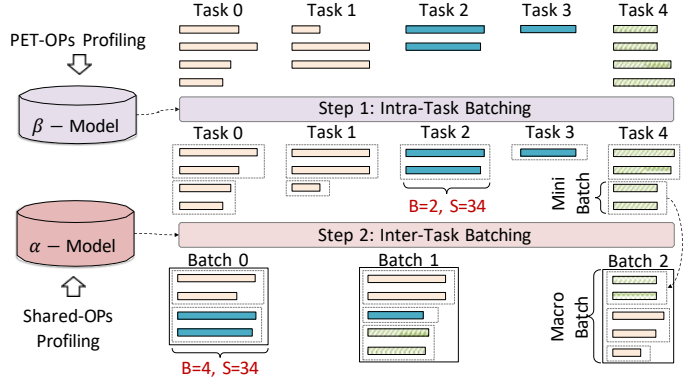


Fig. 13: Coordinated Batching (CB) Strategy

VII. CONCLUSION AND FUTURE DIRECTIONS

In the current era dominated by large models and large datasets, PEFT stands out as a highly attractive method for efficiently adapting models to downstream tasks. This technique gains its appeal by addressing the significant challenges posed by traditional full-model fine-tuning, which often places untenable computational and data demands on average users. This survey provides a systematic review of the latest advancements in PEFT, covering algorithmic developments, computational and efficiency aspects, applications, and system deployment. It offers a comprehensive taxonomy and explanation that serves as an excellent guidance and knowledge base, which enables readers of various levels and disciplines to swiftly grasp the core concepts of PEFT.

For further research on PEFT, we propose a series of possible directions from both algorithm and system perspectives, hoping to inspire more researchers to engage in further studies in these areas.

A. Simplify hyperparameter tuning

The effectiveness of PEFT is often sensitive to its hyperparameters, such as adapters' bottleneck dimension r , LoRA's rank, and the placement of different additive PEFT layers. Manually tuning these hyperparameters will cost lots of efforts. Therefore, future efforts could focus on developing methods that are less dependent on manual tuning of these parameters, or automatically find the optimal hyperparameter settings. Several studies [171], [217], [39], [16], [222], [231] have started to address this issue, but there's a need for more simple and efficient solutions optimizing these hyperparameters.

B. Establish a unified benchmark

Despite the existence of libraries like HuggingFace's PEFT [132] and AdapterHub [147], a comprehensive benchmark for PEFT is still lacking. This gap hinders the ability to fairly compare the performance and efficiency of different PEFT approaches. A well-accepted, up-to-date benchmark akin to MMDetection [17] for object detection would enable researchers to validate their methods against a standard set

of tasks and metrics, fostering innovation and collaboration within the community.

C. Enhance training efficiency

The presumed parameter efficiency of PEFT does not always consistent with computational and memory savings during training. Given that trainable parameters are intertwined within the pre-trained model’s architecture, the full model’s gradients often need to be computed and stored during fine-tuning. This oversight calls for a rethinking of what constitutes efficiency. As outlined in Section IV, potential solutions lie in the integration of model compression techniques such as pruning and quantization, alongside innovations specifically designed to optimize memory during PEFT tuning. Further research into enhancing the computational efficiency of PEFT methodologies is imperative.

D. Explore scaling laws

The design and effectiveness of PEFT methods originally developed for smaller Transformer models do not necessarily scale with larger models. As the size of foundation models increases, identifying and adapting PEFT strategies that remain effective is crucial. This exploration will help tailor PEFT approaches to the evolving landscape of large model architectures.

E. Serve more models and tasks

The rise of large foundation models across various domains presents new opportunities for PEFT. Designing PEFT methods tailored to the unique characteristics of models, such as Sora [9], Mamba [58], and LVM [6], can unlock new application scenarios and opportunities.

F. Enhancing data privacy

Trusting centralized systems to serve or fine-tune personalized PEFT modules is yet another issue for system developers. Side-channel attackers have successfully deployed to reconstruct user’s data by hijacking the intermediate results. One perspective of future trust-worthy LLM system design involves developing an encryption protocol for both personal data and intermediate training and inference results.

G. PEFT with model compression

Model compression is one of the most effective ways to make LLM executable on resource-limited devices. Yet, the impact of model compression techniques on the performance of PEFT algorithms running on hardware remains another systemic challenge. Common compression techniques such as quantization and pruning necessitate dedicated hardware platforms to expedite the process, and building such hardware platform for compressed models is yet another direction for researchers.

REFERENCES

- [1] AGHAJANYAN, A., ZETTLEMOYER, L., AND GUPTA, S. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. *arXiv preprint arXiv:2012.13255* (2020).
- [2] ALAYRAC, J.-B., DONAHUE, J., LUC, P., MIECH, A., BARR, I., HASSON, Y., LENC, K., MENSCH, A., MILLICAN, K., REYNOLDS, M., ET AL. Flamingo: a visual language model for few-shot learning. *Advances in Neural Information Processing Systems* 35 (2022), 23716–23736.
- [3] ANSELL, A., PONTI, E. M., KORHONEN, A., AND VULIĆ, I. Composable sparse fine-tuning for cross-lingual transfer. *arXiv preprint arXiv:2110.07560* (2021).
- [4] ANTOL, S., AGRAWAL, A., LU, J., MITCHELL, M., BATRA, D., ZITNICK, C. L., AND PARIKH, D. Vqa: Visual question answering. In *Proceedings of the IEEE international conference on computer vision* (2015), pp. 2425–2433.
- [5] ANTORÁN, J., JANZ, D., ALLINGHAM, J. U., DAXBERGER, E., BARBANO, R. R., NALISNICK, E., AND HERNÁNDEZ-LOBATO, J. M. Adapting the linearised laplace model evidence for modern deep learning. In *International Conference on Machine Learning* (2022), PMLR, pp. 796–821.
- [6] BAI, Y., GENG, X., MANGALAM, K., BAR, A., YUILLE, A., DARRELL, T., MALIK, J., AND EFROS, A. A. Sequential modeling enables scalable learning for large vision models. *arXiv preprint arXiv:2312.00785* (2023).
- [7] BECK, A., AND TBOULLE, M. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM journal on imaging sciences* 2, 1 (2009), 183–202.
- [8] BISK, Y., ZELLERS, R., BRAS, R. L., GAO, J., AND CHOI, Y. Piqa: Reasoning about physical commonsense in natural language. In *Thirty-Fourth AAAI Conference on Artificial Intelligence* (2020).
- [9] BROOKS, T., PEEBLES, B., HOLMES, C., DEPUÉ, W., GUO, Y., JING, L., SCHNURR, D., TAYLOR, J., LUHMAN, T., LUHMAN, E., NG, C., WANG, R., AND RAMESH, A. Video generation models as world simulators.
- [10] BROWN, T., MANN, B., RYDER, N., SUBBIAH, M., KAPLAN, J. D., DHARIWAL, P., NEELAKANTAN, A., SHYAM, P., SASTRY, G., ASKELL, A., ET AL. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [11] CHAI, W., ZHENG, D., CAO, J., CHEN, Z., WANG, C., AND MA, C. Speedupnet: A plug-and-play hyper-network for accelerating text-to-image diffusion models. *arXiv preprint arXiv:2312.08887* (2023).
- [12] CHAI, Y., GKOUNTOURAS, J., KO, G. G., BROOKS, D., AND WEI, G.-Y. Int2. 1: Towards fine-tunable quantized large language models with error correction through low-rank adaptation. *arXiv preprint arXiv:2306.08162* (2023).
- [13] CHAMBOLE, A., DE VORE, R. A., LEE, N.-Y., AND LUCIER, B. J. Nonlinear wavelet image processing: variational problems, compression, and noise removal through wavelet shrinkage. *IEEE Transactions on image processing* 7, 3 (1998), 319–335.
- [14] CHANG, C.-H., KAYED, M., GIRGIS, M. R., AND SHAALAN, K. F. A survey of web information extraction systems. *IEEE transactions on knowledge and data engineering* 18, 10 (2006), 1411–1428.
- [15] CHE, T., LIU, J., ZHOU, Y., REN, J., ZHOU, J., SHENG, V. S., DAI, H., AND DOU, D. Federated learning of large language models with parameter-efficient prompt tuning and adaptive optimization. *arXiv preprint arXiv:2310.15080* (2023).
- [16] CHEN, J., ZHANG, A., SHI, X., LI, M., SMOLA, A., AND YANG, D. Parameter-efficient fine-tuning design spaces. *arXiv preprint arXiv:2301.01821* (2023).
- [17] CHEN, K., WANG, J., PANG, J., CAO, Y., XIONG, Y., LI, X., SUN, S., FENG, W., LIU, Z., XU, J., ZHANG, Z., CHENG, D., ZHU, C., CHENG, T., ZHAO, Q., LI, B., LU, X., ZHU, R., WU, Y., DAI, J., WANG, J., SHI, J., OUYANG, W., LOY, C. C., AND LIN, D. MMDetection: Open mmlab detection toolbox and benchmark. *arXiv preprint arXiv:1906.07155* (2019).
- [18] CHEN, L., HUANG, H., AND CHENG, M. Ptp: Boosting stability and performance of prompt tuning with perturbation-based regularizer. *arXiv preprint arXiv:2305.02423* (2023).
- [19] CHEN, L., YE, Z., WU, Y., ZHUO, D., CEZE, L., AND KRISHNAMURTHY, A. Punica: Multi-tenant lora serving. *arXiv preprint arXiv:2310.18547* (2023).
- [20] CHEN, M., PENG, H., FU, J., AND LING, H. Autoformer: Searching transformers for visual recognition. In *Proceedings of the IEEE/CVF international conference on computer vision* (2021), pp. 12270–12280.

- [21] CHEN, S., GE, C., TONG, Z., WANG, J., SONG, Y., WANG, J., AND LUO, P. Adaptformer: Adapting vision transformers for scalable visual recognition. *Advances in Neural Information Processing Systems* 35 (2022), 16664–16678.
- [22] CHEN, S., WONG, S., CHEN, L., AND TIAN, Y. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595* (2023).
- [23] CHEN, X., XIE, S., AND HE, K. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (2021), pp. 9640–9649.
- [24] CHEN, Y., QIAN, S., TANG, H., LAI, X., LIU, Z., HAN, S., AND JIA, J. Longlora: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307* (2023).
- [25] CHEN, Z., DUAN, Y., WANG, W., HE, J., LU, T., DAI, J., AND QIAO, Y. Vision transformer adapter for dense predictions. *arXiv preprint arXiv:2205.08534* (2022).
- [26] CHO, J., LEI, J., TAN, H., AND BANSAL, M. Unifying vision-and-language tasks via text generation. In *International Conference on Machine Learning* (2021), PMLR, pp. 1931–1942.
- [27] CHOI, J.-Y., KIM, J., PARK, J.-H., MOK, W.-L., AND LEE, S. Smop: Towards efficient and effective prompt tuning with sparse mixture-of-prompts. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (2023), pp. 14306–14316.
- [28] CHOI, Y., AND LEE, J.-H. Codeprompt: Task-agnostic prefix tuning for program and language generation. In *Findings of the Association for Computational Linguistics: ACL 2023* (2023), pp. 5282–5297.
- [29] CHRONOPOULOU, A., PETERS, M. E., FRASER, A., AND DODGE, J. Adaptersoup: Weight averaging to improve generalization of pretrained language models. *arXiv preprint arXiv:2302.07027* (2023).
- [30] CLARK, C. E. A. Boolq: Exploring the surprising difficulty of natural yes/no questions. In *NAACL* (2019).
- [31] CLARK, P., COWHEY, I., ETZIONI, O., KHOT, T., SABHARWAL, A., SCHOENICK, C., AND TAFJORD, O. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv:1803.05457v1* (2018).
- [32] CROITORU, F.-A., HONDRU, V., IONESCU, R. T., AND SHAH, M. Diffusion models in vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).
- [33] DAI, Y., LANG, H., ZHENG, Y., HUANG, F., SI, L., AND LI, Y. Lifelong learning for question answering with hierarchical prompts. *arXiv preprint arXiv:2208.14602* (2022).
- [34] DAS, S. S., ZHANG, R. H., SHI, P., YIN, W., AND ZHANG, R. Unified low-resource sequence labeling by sample-aware dynamic sparse finetuning. *arXiv preprint arXiv:2311.03748* (2023).
- [35] DEGHANI, M., DJOLONGA, J., MUSTAFA, B., PADLEWSKI, P., HEK, J., GILMER, J., STEINER, A. P., CARON, M., GEIRHOS, R., ALABDULMOHSIN, I., ET AL. Scaling vision transformers to 22 billion parameters. In *International Conference on Machine Learning* (2023), PMLR, pp. 7480–7512.
- [36] DETTMERS, T., LEWIS, M., BELKADA, Y., AND ZETTMLOYER, L. Gpt3. int8 (): 8-bit matrix multiplication for transformers at scale. *Advances in Neural Information Processing Systems* 35 (2022), 30318–30332.
- [37] DETTMERS, T., PAGNONI, A., HOLTZMAN, A., AND ZETTMLOYER, L. Qlora: Efficient finetuning of quantized llms. *arXiv preprint arXiv:2305.14314* (2023).
- [38] DHARIWAL, P., AND NICHOL, A. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems* 34 (2021), 8780–8794.
- [39] DING, N., LV, X., WANG, Q., CHEN, Y., ZHOU, B., LIU, Z., AND SUN, M. Sparse low-rank adaptation of pre-trained language models. *arXiv preprint arXiv:2311.11696* (2023).
- [40] DOSOVITSKIY, A., BEYER, L., KOLESNIKOV, A., WEISSENBORN, D., ZHAI, X., UNTERTHINER, T., DEGHANI, M., MINDERER, M., HEIGOLD, G., GELLY, S., ET AL. An image is worth 16x16 words: Transformers for image recognition at scale. *arxiv* 2020. *arXiv preprint arXiv:2010.11929* (2010).
- [41] DOVEH, S., ARBELLE, A., HARARY, S., SCHWARTZ, E., HERZIG, R., GIRYES, R., FERIS, R., PANDA, R., ULLMAN, S., AND KARLINSKY, L. Teaching structured vision & language concepts to vision & language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2023), pp. 2657–2668.
- [42] EDALATI, A., TAHAIEI, M., KOBYZEV, I., NIA, V. P., CLARK, J. J., AND REZAGHOLIZADEH, M. Krona: Parameter efficient tuning with kronecker adapter. *arXiv preprint arXiv:2212.10650* (2022).
- [43] EVERINGHAM, M., VAN GOOL, L., WILLIAMS, C. K., WINN, J., AND ZISSERMAN, A. The pascal visual object classes (voc) challenge. *International journal of computer vision* 88 (2010), 303–338.
- [44] FENG, C.-M., YU, K., LIU, Y., KHAN, S., AND ZUO, W. Diverse data augmentation with diffusions for effective test-time prompt tuning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 2704–2714.
- [45] FENG, W., HAO, C., ZHANG, Y., HAN, Y., AND WANG, H. Mixture-of-loras: An efficient multitask tuning for large language models. *arXiv preprint arXiv:2403.03432* (2024).
- [46] FRANKLE, J., AND CARBIN, M. The lottery ticket hypothesis: Finding sparse, trainable neural networks. *arXiv preprint arXiv:1803.03635* (2018).
- [47] FRAZIER, P. I. A tutorial on bayesian optimization. *arXiv preprint arXiv:1807.02811* (2018).
- [48] FU, Z., YANG, H., SO, A. M.-C., LAM, W., BING, L., AND COLLIER, N. On the effectiveness of parameter-efficient fine-tuning. In *Proceedings of the AAAI Conference on Artificial Intelligence* (2023), vol. 37, pp. 12799–12807.
- [49] GAL, R., ALALUF, Y., ATZMON, Y., PATASHNIK, O., BERMANO, A. H., CHECHIK, G., AND COHEN-OR, D. An image is worth one word: Personalizing text-to-image generation using textual inversion. *arXiv preprint arXiv:2208.01618* (2022).
- [50] GANDIKOTA, R., MATERZYNSKA, J., ZHOU, T., TORRALBA, A., AND BAU, D. Concept sliders: Lora adaptors for precise control in diffusion models. *arXiv preprint arXiv:2311.12092* (2023).
- [51] GAO, P., GENG, S., ZHANG, R., MA, T., FANG, R., ZHANG, Y., LI, H., AND QIAO, Y. Clip-adapter: Better vision-language models with feature adapters. *International Journal of Computer Vision* (2023), 1–15.
- [52] GAO, P., HAN, J., ZHANG, R., LIN, Z., GENG, S., ZHOU, A., ZHANG, W., LU, P., HE, C., YUE, X., ET AL. Llama-adapter v2: Parameter-efficient visual instruction model. *arXiv preprint arXiv:2304.15010* (2023).
- [53] GAO, Q., ZHAO, C., SUN, Y., XI, T., ZHANG, G., GHANEM, B., AND ZHANG, J. A unified continual learning framework with general parameter-efficient tuning. *arXiv preprint arXiv:2303.10070* (2023).
- [54] GHEINI, M., REN, X., AND MAY, J. Cross-attention is all you need: Adapting pretrained transformers for machine translation. *arXiv preprint arXiv:2104.08771* (2021).
- [55] GOMEZ, A. N., REN, M., URTASUN, R., AND GROSSE, R. B. The reversible residual network: Backpropagation without storing activations. *Advances in neural information processing systems* 30 (2017).
- [56] GOYAL, R., EBRAHIMI KAHOU, S., MICHALSKI, V., MATERZYNSKA, J., WESTPHAL, S., KIM, H., HAENEL, V., FRUEND, I., YIANNILOS, P., MUELLER-FREITAG, M., ET AL. The “something” video database for learning and evaluating visual common sense. In *Proceedings of the IEEE international conference on computer vision* (2017), pp. 5842–5850.
- [57] GRAVE, E., CISSE, M. M., AND JOULIN, A. Unbounded cache model for online language modeling with open vocabulary. *Advances in neural information processing systems* 30 (2017).
- [58] GU, A., AND DAO, T. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752* (2023).
- [59] GUO, D., RUSH, A. M., AND KIM, Y. Parameter-efficient transfer learning with diff pruning. *arXiv preprint arXiv:2012.07463* (2020).
- [60] GUO, H., GREENGARD, P., XING, E. P., AND KIM, Y. Lq-lora: Low-rank plus quantized matrix decomposition for efficient language model finetuning. *arXiv preprint arXiv:2311.12023* (2023).
- [61] HADI, M. U., QURESHI, R., SHAH, A., IRFAN, M., ZAFAR, A., SHAIKH, M., AKHTAR, N., WU, J., AND MIRJALILI, S. A survey on large language models: Applications, challenges, limitations, and practical usage. *TechRxiv* (2023).
- [62] HAN, Z., WANG, Y., ZHOU, L., WANG, P., YAN, B., ZHOU, J., WANG, Y., AND SHEN, D. Contrastive diffusion model with auxiliary guidance for coarse-to-fine pet reconstruction. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (2023), Springer, pp. 239–249.
- [63] HAN, Z., ZHU, F., LAO, Q., AND JIANG, H. Zero-shot referring expression comprehension via structural similarity between images and captions. *arXiv preprint arXiv:2311.17048* (2023).
- [64] HAOBO, S., ZHAO, H., MAJUMDER, S., AND LIN, T. Increasing model capacity for free: A simple strategy for parameter efficient fine-tuning. In *The Twelfth International Conference on Learning Representations* (2023).
- [65] HAYOU, S., GHOSH, N., AND YU, B. Lora+: Efficient low rank adaptation of large models. *arXiv preprint arXiv:2402.12354* (2024).

- [66] HE, H., CAI, J., ZHANG, J., TAO, D., AND ZHUANG, B. Sensitivity-aware visual parameter-efficient fine-tuning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 11825–11835.
- [67] HE, J., ZHOU, C., MA, X., BERG-KIRKPATRICK, T., AND NEUBIG, G. Towards a unified view of parameter-efficient transfer learning. *arXiv preprint arXiv:2110.04366* (2021).
- [68] HE, K., CHEN, X., XIE, S., LI, Y., DOLLÁR, P., AND GIRSHICK, R. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (2022), pp. 16000–16009.
- [69] HE, S., DING, L., DONG, D., ZHANG, J., AND TAO, D. SparseAdapter: An easy approach for improving the parameter-efficiency of adapters. In *Findings of the Association for Computational Linguistics: EMNLP 2022* (Abu Dhabi, United Arab Emirates, Dec. 2022), Association for Computational Linguistics, pp. 2184–2190.
- [70] HE, S., FAN, R.-Z., DING, L., SHEN, L., ZHOU, T., AND TAO, D. Mera: Merging pretrained adapters for few-shot learning. *arXiv preprint arXiv:2308.15982* (2023).
- [71] HE, X., LI, C., ZHANG, P., YANG, J., AND WANG, X. E. Parameter-efficient model adaptation for vision transformers. In *Proceedings of the AAAI Conference on Artificial Intelligence* (2023), vol. 37, pp. 817–825.
- [72] HEDEGAARD, L., ALOK, A., JOSE, J., AND IOSIFIDIS, A. Structured pruning adapters. *arXiv preprint arXiv:2211.10155* (2022).
- [73] HO, J., JAIN, A., AND ABBEEL, P. Denoising diffusion probabilistic models. *Advances in neural information processing systems* 33 (2020), 6840–6851.
- [74] HOSSAIN, M. Z., SOHEL, F., SHIRATUDDIN, M. F., AND LAGA, H. A comprehensive survey of deep learning for image captioning. *ACM Computing Surveys (CSUR)* 51, 6 (2019), 1–36.
- [75] HOULSBY, N., GIURGIU, A., JASTRZEBSKI, S., MORRONE, B., DE LAROUSSILHE, Q., GESMUNDO, A., ATTARIYAN, M., AND GELLY, S. Parameter-efficient transfer learning for nlp. In *International Conference on Machine Learning* (2019), PMLR, pp. 2790–2799.
- [76] HU, E. J., SHEN, Y., WALLIS, P., ALLEN-ZHU, Z., LI, Y., WANG, S., WANG, L., AND CHEN, W. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685* (2021).
- [77] HU, S., ZHANG, Z., DING, N., WANG, Y., WANG, Y., LIU, Z., AND SUN, M. Sparse structure search for parameter-efficient tuning. *arXiv preprint arXiv:2206.07382* (2022).
- [78] HU, Z., LAN, Y., WANG, L., XU, W., LIM, E.-P., LEE, R. K.-W., BING, L., AND PORIA, S. Llm-adapters: An adapter family for parameter-efficient fine-tuning of large language models. *arXiv preprint arXiv:2304.01933* (2023).
- [79] HU, Z.-Y., LI, Y., LYU, M. R., AND WANG, L. Vl-pet: Vision-and-language parameter-efficient tuning via granularity control. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 3010–3020.
- [80] HUANG, C., LIU, Q., LIN, B. Y., PANG, T., DU, C., AND LIN, M. LoraHub: Efficient cross-task generalization via dynamic lora composition. *arXiv preprint arXiv:2307.13269* (2023).
- [81] HUANG, T., DONG, B., YANG, Y., HUANG, X., LAU, R. W., OUYANG, W., AND ZUO, W. Clip2point: Transfer clip to point cloud classification with image-depth pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 22157–22167.
- [82] HUANG, Y., LI, Y., XU, Y., ZHANG, L., GAN, R., ZHANG, J., AND WANG, L. Mvp-tuning: Multi-view knowledge retrieval with prompt tuning for commonsense reasoning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023), pp. 13417–13432.
- [83] JIA, C., YANG, Y., XIA, Y., CHEN, Y.-T., PAREKH, Z., PHAM, H., LE, Q., SUNG, Y.-H., LI, Z., AND DUEK, T. Scaling up visual and vision-language representation learning with noisy text supervision. In *International conference on machine learning* (2021), PMLR, pp. 4904–4916.
- [84] JIA, M., TANG, L., CHEN, B.-C., CARDIE, C., BELONGIE, S., HARIHARAN, B., AND LIM, S.-N. Visual prompt tuning. In *European Conference on Computer Vision* (2022), Springer, pp. 709–727.
- [85] JIANG, Z., MAO, C., HUANG, Z., MA, A., LV, Y., SHEN, Y., ZHAO, D., AND ZHOU, J. Res-tuning: A flexible and efficient tuning paradigm via unbinding tuner from backbone. *arXiv preprint arXiv:2310.19859* (2023).
- [86] JIE, S., AND DENG, Z.-H. Convolutional bypasses are better vision transformer adapters. *arXiv preprint arXiv:2207.07039* (2022).
- [87] JIE, S., WANG, H., AND DENG, Z.-H. Revisiting the parameter efficiency of adapters from the perspective of precision redundancy. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 17217–17226.
- [88] JIN, F., ZHANG, J., AND ZONG, C. Parameter-efficient tuning for large language model without calculating its gradients. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (2023), pp. 321–330.
- [89] JU, C., HAN, T., ZHENG, K., ZHANG, Y., AND XIE, W. Prompting visual-language models for efficient video understanding. In *European Conference on Computer Vision* (2022), Springer, pp. 105–124.
- [90] KANG, H., ZHANG, Q., KUNDU, S., JEONG, G., LIU, Z., KRISHNA, T., AND ZHAO, T. Gear: An efficient kv cache compression recipe for near-lossless generative inference of llm. *arXiv preprint arXiv:2403.05527* (2024).
- [91] KARIMI MAHABADI, R., HENDERSON, J., AND RUDER, S. Compacter: Efficient low-rank hypercomplex adapter layers. *Advances in Neural Information Processing Systems* 34 (2021), 1022–1035.
- [92] KAY, W., CARREIRA, J., SIMONYAN, K., ZHANG, B., HILLIER, C., VIJAYANARASIMHAN, S., VIOLA, F., GREEN, T., BACK, T., NATSEV, P., ET AL. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950* (2017).
- [93] KHATTAK, M. U., RASHEED, H., MAAZ, M., KHAN, S., AND KHAN, F. S. Maple: Multi-modal prompt learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2023), pp. 19113–19122.
- [94] KIM, J., LEE, J. H., KIM, S., PARK, J., YOO, K. M., KWON, S. J., AND LEE, D. Memory-efficient fine-tuning of compressed large language models via sub-4-bit integer quantization. *arXiv preprint arXiv:2305.14152* (2023).
- [95] KIRKPATRICK, J., PASCANU, R., RABINOWITZ, N., VENESS, J., DESJARDINS, G., RUSU, A. A., MILAN, K., QUAN, J., RAMALHO, T., GRABSKA-BARWINSKA, A., ET AL. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences* 114, 13 (2017), 3521–3526.
- [96] KUEHNE, H., JHUANG, H., GARROTE, E., POGGIO, T., AND SERRE, T. Hmdb: a large video database for human motion recognition. In *2011 International conference on computer vision* (2011), IEEE, pp. 2556–2563.
- [97] KUMARI, N., ZHANG, B., ZHANG, R., SHECHTMAN, E., AND ZHU, J.-Y. Multi-concept customization of text-to-image diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2023), pp. 1931–1941.
- [98] KWON, W., LI, Z., ZHUANG, S., SHENG, Y., ZHENG, L., YU, C. H., GONZALEZ, J., ZHANG, H., AND STOICA, I. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles* (2023), pp. 611–626.
- [99] LAWTON, N., KUMAR, A., THATTAI, G., GALSTYAN, A., AND STEEG, G. V. Neural architecture search for parameter-efficient fine-tuning of large pre-trained language models. *arXiv preprint arXiv:2305.16597* (2023).
- [100] LEE, S. Toward continual learning for conversational agents. *arXiv preprint arXiv:1712.09943* (2017).
- [101] LEI, T., BAI, J., BRAHMA, S., AINSLIE, J., LEE, K., ZHOU, Y., DU, N., ZHAO, V. Y., WU, Y., LI, B., ET AL. Conditional adapters: Parameter-efficient transfer learning with fast inference. *arXiv preprint arXiv:2304.04947* (2023).
- [102] LESTER, B., AL-ROU, R., AND CONSTANT, N. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691* (2021).
- [103] LI, G., HAMMOUD, H. A. A. K., ITANI, H., KHIZBULLIN, D., AND GHANEM, B. Camel: Communicative agents for “mind” exploration of large language model society. In *Thirty-seventh Conference on Neural Information Processing Systems* (2023).
- [104] LI, J., AITKEN, W., BHAMBHORIA, R., AND ZHU, X. Prefix propagation: Parameter-efficient tuning for long sequences. *arXiv preprint arXiv:2305.12086* (2023).
- [105] LI, X. L., AND LIANG, P. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190* (2021).
- [106] LI, Y., DU, M., WANG, X., AND WANG, Y. Prompt tuning pushes farther, contrastive learning pulls closer: A two-stage approach to mitigate social biases. *arXiv preprint arXiv:2307.01595* (2023).
- [107] LI, Y., LIANG, F., ZHAO, L., CUI, Y., OUYANG, W., SHAO, J., YU, F., AND YAN, J. Supervision exists everywhere: A data efficient contrastive language-image pre-training paradigm. *arXiv preprint arXiv:2110.05208* (2021).

- [108] LI, Y., YU, Y., LIANG, C., HE, P., KARAMPATZIAKIS, N., CHEN, W., AND ZHAO, T. Loftq: Lora-fine-tuning-aware quantization for large language models. *arXiv preprint arXiv:2310.08659* (2023).
- [109] LIAN, D., ZHOU, D., FENG, J., AND WANG, X. Scaling & shifting your features: A new baseline for efficient model tuning. *Advances in Neural Information Processing Systems* 35 (2022), 109–123.
- [110] LIANG, Z., WEI, F., JIE, Y., QIAN, Y., HAO, Z., AND HAN, B. Prompts can play lottery tickets well: Achieving lifelong information extraction via lottery prompt tuning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023), pp. 277–292.
- [111] LIAO, B., MENG, Y., AND MONZ, C. Parameter-efficient fine-tuning without introducing new latency. *arXiv preprint arXiv:2305.16742* (2023).
- [112] LIAO, B., TAN, S., AND MONZ, C. Make your pre-trained model reversible: From parameter to memory efficient fine-tuning. *arXiv preprint arXiv:2306.00477* (2023).
- [113] LIN, T.-Y., MAIRE, M., BELONGIE, S., HAYS, J., PERONA, P., RAMANAN, D., DOLLÁR, P., AND ZITNICK, C. L. Microsoft coco: Common objects in context. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13* (2014), Springer, pp. 740–755.
- [114] LIN, Z., GENG, S., ZHANG, R., GAO, P., DE MELO, G., WANG, X., DAI, J., QIAO, Y., AND LI, H. Frozen clip models are efficient video learners. In *European Conference on Computer Vision* (2022), Springer, pp. 388–404.
- [115] LIU, H., LI, C., WU, Q., AND LEE, Y. J. Visual instruction tuning. *arXiv preprint arXiv:2304.08485* (2023).
- [116] LIU, H., TAM, D., MUQEETH, M., MOHTA, J., HUANG, T., BANSAL, M., AND RAFFEL, C. A. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. *Advances in Neural Information Processing Systems* 35 (2022), 1950–1965.
- [117] LIU, J., MOREAU, A., PREUSS, M., RAPIN, J., ROZIERE, B., TEYTAUD, F., AND TEYTAUD, O. Versatile black-box optimization. In *Proceedings of the 2020 Genetic and Evolutionary Computation Conference* (2020), pp. 620–628.
- [118] LIU, J., XIAO, G., LI, K., LEE, J. D., HAN, S., DAO, T., AND CAI, T. Bitdelta: Your fine-tune may only be worth one bit. *arXiv preprint arXiv:2402.10193* (2024).
- [119] LIU, Q., WU, X., ZHAO, X., ZHU, Y., XU, D., TIAN, F., AND ZHENG, Y. Moelora: An moe-based parameter efficient fine-tuning method for multi-task medical applications. *arXiv preprint arXiv:2310.18339* (2023).
- [120] LIU, S.-Y., WANG, C.-Y., YIN, H., MOLCHANOV, P., WANG, Y.-C. F., CHENG, K.-T., AND CHEN, M.-H. Dora: Weight-decomposed low-rank adaptation. *arXiv preprint arXiv:2402.09353* (2024).
- [121] LIU, X., JI, K., FU, Y., TAM, W. L., DU, Z., YANG, Z., AND TANG, J. P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks. *arXiv preprint arXiv:2110.07602* (2021).
- [122] LIU, X., SUN, T., HUANG, X., AND QIU, X. Late prompt tuning: A late prompt could be better than many prompts. *arXiv preprint arXiv:2210.11292* (2022).
- [123] LIU, X., ZHENG, Y., DU, Z., DING, M., QIAN, Y., YANG, Z., AND TANG, J. Gpt understands, too. *arXiv preprint arXiv:2103.10385* (2021).
- [124] LU, X., BRAHMAN, F., WEST, P., JANG, J., CHANDU, K., RAVICHANDER, A., QIN, L., AMMANABROLU, P., JIANG, L., RAMNATH, S., ET AL. Inference-time policy adapters (ipa): Tailoring extreme-scale lms without fine-tuning. *arXiv preprint arXiv:2305.15065* (2023).
- [125] LUO, S., TAN, Y., PATIL, S., GU, D., VON PLATEN, P., PASSOS, A., HUANG, L., LI, J., AND ZHAO, H. Lcm-lora: A universal stable-diffusion acceleration module. *arXiv preprint arXiv:2311.05556* (2023).
- [126] MA, F., ZHANG, C., REN, L., WANG, J., WANG, Q., WU, W., QUAN, X., AND SONG, D. Xprompt: Exploring the extreme of prompt tuning. *arXiv preprint arXiv:2210.04457* (2022).
- [127] MACKAY, D. J. A practical bayesian framework for backpropagation networks. *Neural computation* 4, 3 (1992), 448–472.
- [128] MADOTTO, A., LIN, Z., ZHOU, Z., MOON, S., CROOK, P., LIU, B., YU, Z., CHO, E., AND WANG, Z. Continual learning in task-oriented dialogue systems. *arXiv preprint arXiv:2012.15504* (2020).
- [129] MAHABADI, R. K., RUDER, S., DEGHANI, M., AND HENDERSON, J. Parameter-efficient multi-task fine-tuning for transformers via shared hypernetworks. *arXiv preprint arXiv:2106.04489* (2021).
- [130] MALACH, E., YEHUDAI, G., SHALEV-SCHWARTZ, S., AND SHAMIR, O. Proving the lottery ticket hypothesis: Pruning is all you need. In *International Conference on Machine Learning* (2020), PMLR, pp. 6682–6691.
- [131] MALLADI, S., GAO, T., NICHANI, E., DAMIAN, A., LEE, J. D., CHEN, D., AND ARORA, S. Fine-tuning language models with just forward passes. *arXiv preprint arXiv:2305.17333* (2023).
- [132] MANGRULKAR, S., GUGGER, S., DEBUT, L., BELKADA, Y., PAUL, S., AND BOSSAN, B. Peft: State-of-the-art parameter-efficient fine-tuning methods. <https://github.com/huggingface/peft>, 2022.
- [133] MAO, Y., MATHIAS, L., HOU, R., ALMAHAIRI, A., MA, H., HAN, J., YIH, W.-T., AND KHABSA, M. Unipelt: A unified framework for parameter-efficient language model tuning. *arXiv preprint arXiv:2110.07577* (2021).
- [134] MENG, X., DAI, D., LUO, W., YANG, Z., WU, S., WANG, X., WANG, P., DONG, Q., CHEN, L., AND SUI, Z. Periodiclora: Breaking the low-rank bottleneck in lora optimization. *arXiv preprint arXiv:2402.16141* (2024).
- [135] MIHAYLOV, T., CLARK, P., KHOT, T., AND SABHARWAL, A. Can a suit of armor conduct electricity? a new dataset for open book question answering. In *EMNLP* (2018).
- [136] MOU, C., WANG, X., XIE, L., WU, Y., ZHANG, J., QI, Z., SHAN, Y., AND QIE, X. T2i-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models. *arXiv preprint arXiv:2302.08453* (2023).
- [137] NAG, S., ZHU, X., SONG, Y.-Z., AND XIANG, T. Zero-shot temporal action detection via vision-language prompting. In *European Conference on Computer Vision* (2022), Springer, pp. 681–697.
- [138] NI, B., PENG, H., CHEN, M., ZHANG, S., MENG, G., FU, J., XIANG, S., AND LING, H. Expanding language-image pretrained models for general video recognition. In *European Conference on Computer Vision* (2022), Springer, pp. 1–18.
- [139] OPENAI. Gpt-4. In <https://openai.com/gpt-4> (2023).
- [140] ORHAN, E. A simple cache model for image recognition. *Advances in Neural Information Processing Systems* 31 (2018).
- [141] OUYANG, Y., CAO, Y., GAO, Y., WU, Z., ZHANG, J., AND DAI, X. On prefix-tuning for lightweight out-of-distribution detection. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023), pp. 1533–1545.
- [142] OZDAYI, M. S., PERIS, C., FITZGERALD, J., DUPUY, C., MAJUMDAR, J., KHAN, H., PARIKH, R., AND GUPTA, R. Controlling the extraction of memorized data from large language models via prompt-tuning. *arXiv preprint arXiv:2305.11759* (2023).
- [143] PAN, J., LIN, Z., ZHU, X., SHAO, J., AND LI, H. St-adapter: Parameter-efficient image-to-video transfer learning. *Advances in Neural Information Processing Systems* 35 (2022), 26462–26477.
- [144] PETROV, A., TORR, P. H., AND BIBI, A. When do prompting and prefix-tuning work? a theory of capabilities and limitations. *arXiv preprint arXiv:2310.19698* (2023).
- [145] PFEIFFER, J., KAMATH, A., RÜCKLÉ, A., CHO, K., AND GUREVYCH, I. Adapterfusion: Non-destructive task composition for transfer learning. *arXiv preprint arXiv:2005.00247* (2020).
- [146] PHANG, J., MAO, Y., HE, P., AND CHEN, W. Hypertuning: Toward adapting large language models without back-propagation. In *International Conference on Machine Learning* (2023), PMLR, pp. 27854–27875.
- [147] POTH, C., STERZ, H., PAUL, I., PURKAYASTHA, S., ENGLÄNDER, L., IMHOF, T., VULIĆ, I., RUDER, S., GUREVYCH, I., AND PFEIFFER, J. Adapters: A unified library for parameter-efficient and modular transfer learning, 2023.
- [148] QIN, Y., WANG, X., SU, Y., LIN, Y., DING, N., YI, J., CHEN, W., LIU, Z., LI, J., HOU, L., ET AL. Exploring universal intrinsic task subspace via prompt tuning. *arXiv preprint arXiv:2110.07867* (2021).
- [149] RADFORD, A., KIM, J. W., HALLACY, C., RAMESH, A., GOH, G., AGARWAL, S., SASTRY, G., ASKELL, A., MISHKIN, P., CLARK, J., ET AL. Learning transferable visual models from natural language supervision. In *International conference on machine learning* (2021), PMLR, pp. 8748–8763.
- [150] RAJABZADEH, H., VALIPOUR, M., ZHU, T., TAHAIE, M., KWON, H. J., GHODSI, A., CHEN, B., AND REZAGHOLIZADEH, M. Qdylora: Quantized dynamic low-rank adaptation for efficient large language model tuning. *arXiv preprint arXiv:2402.10462* (2024).
- [151] REN, L., CHEN, C., WANG, L., AND HUA, K. Learning semantic proxies from visual prompts for parameter-efficient fine-tuning in deep metric learning. *arXiv preprint arXiv:2402.02340* (2024).

- [152] RENNIE, S. J., MARCHERET, E., MROUEH, Y., ROSS, J., AND GOEL, V. Self-critical sequence training for image captioning. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (2017), pp. 7008–7024.
- [153] ROMBACH, R., BLATTMANN, A., LORENZ, D., ESSER, P., AND OMMER, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (2022), pp. 10684–10695.
- [154] RÜCKLÉ, A., GEIGLE, G., GLOCKNER, M., BECK, T., PFEIFFER, J., REIMERS, N., AND GUREVYCH, I. Adapterdrop: On the efficiency of adapters in transformers. *arXiv preprint arXiv:2010.11918* (2020).
- [155] RUIZ, N., LI, Y., JAMPANI, V., PRITCH, Y., RUBINSTEIN, M., AND ABERMAN, K. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2023), pp. 22500–22510.
- [156] SAKAGUCHI, K., BRAS, R. L., BHAGAVATULA, C., AND CHOI, Y. Winogrande: An adversarial winograd schema challenge at scale. *Communications of the ACM* 64, 9 (2021), 99–106.
- [157] SAP, M., RASHKIN, H., CHEN, D., LEBRAS, R., AND CHOI, Y. Socialliqa: Commonsense reasoning about social interactions. *arXiv preprint arXiv:1904.09728* (2019).
- [158] SHENG, Y., CAO, S., LI, D., HOOPER, C., LEE, N., YANG, S., CHOU, C., ZHU, B., ZHENG, L., KEUTZER, K., ET AL. S-lora: Serving thousands of concurrent lora adapters. *arXiv preprint arXiv:2311.03285* (2023).
- [159] SHENG, Y., ZHENG, L., YUAN, B., LI, Z., RYABININ, M., CHEN, B., LIANG, P., RÉ, C., STOICA, I., AND ZHANG, C. Flexgen: High-throughput generative inference of large language models with a single gpu. In *International Conference on Machine Learning* (2023), PMLR, pp. 31094–31116.
- [160] SHI, Z., AND LIPANI, A. Dept: Decomposed prompt tuning for parameter-efficient fine-tuning. *arXiv preprint arXiv:2309.05173* (2023).
- [161] SHU, M., NIE, W., HUANG, D.-A., YU, Z., GOLDSTEIN, T., ANAND-KUMAR, A., AND XIAO, C. Test-time prompt tuning for zero-shot generalization in vision-language models. *Advances in Neural Information Processing Systems* 35 (2022), 14274–14289.
- [162] SINGH, A., HU, R., GOSWAMI, V., COUAIRON, G., GALUBA, W., ROHRBACH, M., AND KIELA, D. Flava: A foundational language and vision alignment model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022), pp. 15638–15650.
- [163] SOHL-DICKSTEIN, J., WEISS, E., MAHESWARANATHAN, N., AND GANGULI, S. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning* (2015), PMLR, pp. 2256–2265.
- [164] STEINER, A., KOLESNIKOV, A., ZHAI, X., WIGHTMAN, R., USZKOREIT, J., AND BEYER, L. How to train your vit? data, augmentation, and regularization in vision transformers. *arXiv preprint arXiv:2106.10270* (2021).
- [165] SU, J., LU, Y., PAN, S., MURTAHA, A., WEN, B., AND LIU, Y. Roformer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864* (2021).
- [166] SU, Y., WANG, X., QIN, Y., CHAN, C.-M., LIN, Y., WANG, H., WEN, K., LIU, Z., LI, P., LI, J., ET AL. On transferability of prompt tuning for natural language processing. *arXiv preprint arXiv:2111.06719* (2021).
- [167] SUNG, Y.-L., CHO, J., AND BANSAL, M. Lst: Ladder side-tuning for parameter and memory efficient transfer learning. *Advances in Neural Information Processing Systems* 35 (2022), 12991–13005.
- [168] SUNG, Y.-L., CHO, J., AND BANSAL, M. Vl-adapter: Parameter-efficient transfer learning for vision-and-language tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022), pp. 5227–5237.
- [169] SUNG, Y.-L., NAIR, V., AND RAFFEL, C. A. Training neural networks with fixed sparse masks. *Advances in Neural Information Processing Systems* 34 (2021), 24193–24205.
- [170] TEAM, G., ANIL, R., BORGEAUD, S., WU, Y., ALAYRAC, J.-B., YU, J., SORICUT, R., SCHALKWYK, J., DAI, A. M., HAUTH, A., ET AL. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805* (2023).
- [171] VALIPOUR, M., REZAGHOLIZADEH, M., KOBYZEV, I., AND GHODSI, A. Dylora: Parameter efficient tuning of pre-trained models using dynamic search-free low-rank adaptation. *arXiv preprint arXiv:2210.07558* (2022).
- [172] VINIYALS, O., TOSHEV, A., BENGIO, S., AND ERHAN, D. Show and tell: Lessons learned from the 2015 mscoco image captioning challenge. *IEEE transactions on pattern analysis and machine intelligence* 39, 4 (2016), 652–663.
- [173] VU, T., LESTER, B., CONSTANT, N., AL-RFOU, R., AND CER, D. Spot: Better frozen model adaptation through soft prompt transfer. *arXiv preprint arXiv:2110.07904* (2021).
- [174] VUCETIC, D., TAYARANIAN, M., ZIAEEFARD, M., CLARK, J. J., MEYER, B. H., AND GROSS, W. J. Efficient fine-tuning of bert models on the edge. In *2022 IEEE International Symposium on Circuits and Systems (ISCAS)* (2022), IEEE, pp. 1838–1842.
- [175] WANG, A., SINGH, A., MICHAEL, J., HILL, F., LEVY, O., AND BOWMAN, S. R. Glue: A multi-task benchmark and analysis platform for natural language understanding. *arXiv preprint arXiv:1804.07461* (2018).
- [176] WANG, P., WU, Q., SHEN, C., DICK, A., AND VAN DEN HENGEL, A. Fvqa: Fact-based visual question answering. *IEEE transactions on pattern analysis and machine intelligence* 40, 10 (2017), 2413–2427.
- [177] WANG, Q., MAO, Y., WANG, J., YU, H., NIE, S., WANG, S., FENG, F., HUANG, L., QUAN, X., XU, Z., ET AL. Aprompt: Attention prompt tuning for efficient adaptation of pre-trained language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (2023), pp. 9147–9160.
- [178] WANG, X., CHEN, T., GE, Q., XIA, H., BAO, R., ZHENG, R., ZHANG, Q., GUI, T., AND HUANG, X. Orthogonal subspace learning for language model continual learning. *arXiv preprint arXiv:2310.14152* (2023).
- [179] WANG, Y., CHAUHAN, J., WANG, W., AND HSIEH, C.-J. Universality and limitations of prompt tuning. *arXiv preprint arXiv:2305.18787* (2023).
- [180] WANG, Y., MUKHERJEE, S., LIU, X., GAO, J., AWADALLAH, A. H., AND GAO, J. Adamix: Mixture-of-adapter for parameter-efficient tuning of large language models. *arXiv preprint arXiv:2205.12410* 1, 2 (2022), 4.
- [181] WANG, Z., YU, X., RAO, Y., ZHOU, J., AND LU, J. P2p: Tuning pre-trained image models for point cloud analysis with point-to-pixel prompting. *Advances in neural information processing systems* 35 (2022), 14388–14402.
- [182] WANG, Z., ZHANG, Z., LEE, C.-Y., ZHANG, H., SUN, R., REN, X., SU, G., PEROT, V., DY, J., AND PEISTER, T. Learning to prompt for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022), pp. 139–149.
- [183] WRIGHT, J., GANESH, A., RAO, S., PENG, Y., AND MA, Y. Robust principal component analysis: Exact recovery of corrupted low-rank matrices via convex optimization. *Advances in neural information processing systems* 22 (2009).
- [184] WU, H., AND SHI, X. Adversarial soft prompt tuning for cross-domain sentiment analysis. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2022), pp. 2438–2447.
- [185] WU, J., YU, T., WANG, R., SONG, Z., ZHANG, R., ZHAO, H., LU, C., LI, S., AND HENAO, R. Infoprompt: Information-theoretic soft prompt tuning for natural language understanding. *arXiv preprint arXiv:2306.04933* (2023).
- [186] WU, J. Z., GE, Y., WANG, X., LEI, S. W., GU, Y., SHI, Y., HSU, W., SHAN, Y., QIE, X., AND SHOU, M. Z. Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 7623–7633.
- [187] WU, Q., BANSAL, G., ZHANG, J., WU, Y., ZHANG, S., ZHU, E., LI, B., JIANG, L., ZHANG, X., AND WANG, C. Autogen: Enabling next-gen llm applications via multi-agent conversation framework. *arXiv preprint arXiv:2308.08155* (2023).
- [188] WU, Q., TENNEY, D., WANG, P., SHEN, C., DICK, A., AND VAN DEN HENGEL, A. Visual question answering: A survey of methods and datasets. *Computer Vision and Image Understanding* 163 (2017), 21–40.
- [189] WU, Z., WANG, S., GU, J., HOU, R., DONG, Y., VYDISWARAN, V., AND MA, H. Idpg: An instance-dependent prompt generation method. *arXiv preprint arXiv:2204.04497* (2022).
- [190] XIAO, G., LIN, J., AND HAN, S. Offsite-tuning: Transfer learning without full model. *arXiv preprint arXiv:2302.04870* (2023).
- [191] XING, Z., DAI, Q., HU, H., WU, Z., AND JIANG, Y.-G. Simda: Simple diffusion adapter for efficient video generation. *arXiv preprint arXiv:2308.09710* (2023).
- [192] XU, B., LIU, X., SHEN, H., HAN, Z., LI, Y., YUE, M., PENG, Z., LIU, Y., YAO, Z., AND XU, D. Gentopia: A collaborative platform for tool-augmented llms. *arXiv preprint arXiv:2308.04030* (2023).

- [193] XU, M., ZHANG, Z., WEI, F., HU, H., AND BAI, X. Side adapter network for open-vocabulary semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2023), pp. 2945–2954.
- [194] XU, R., LUO, F., ZHANG, Z., TAN, C., CHANG, B., HUANG, S., AND HUANG, F. Raise a child in large language model: Towards effective and generalizable fine-tuning. *arXiv preprint arXiv:2109.05687* (2021).
- [195] XU, Y., XIE, L., GU, X., CHEN, X., CHANG, H., ZHANG, H., CHEN, Z., ZHANG, X., AND TIAN, Q. Qa-lora: Quantization-aware low-rank adaptation of large language models. *arXiv preprint arXiv:2309.14717* (2023).
- [196] XU, Z., CHEN, Z., ZHANG, Y., SONG, Y., WAN, X., AND LI, G. Bridging vision and language encoders: Parameter-efficient tuning for referring image segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 17503–17512.
- [197] YANG, A. X., ROBEYNS, M., WANG, X., AND AITCHISON, L. Bayesian low-rank adaptation for large language models. *arXiv preprint arXiv:2308.13111* (2023).
- [198] YANG, J. Longqlora: Efficient and effective method to extend context length of large language models. *arXiv preprint arXiv:2311.04879* (2023).
- [199] YANG, L., ZHANG, Z., SONG, Y., HONG, S., XU, R., ZHAO, Y., ZHANG, W., CUI, B., AND YANG, M.-H. Diffusion models: A comprehensive survey of methods and applications. *ACM Computing Surveys* 56, 4 (2023), 1–39.
- [200] YANG, T., ZHU, Y., XIE, Y., ZHANG, A., CHEN, C., AND LI, M. Aim: Adapting image models for efficient video action recognition. *arXiv preprint arXiv:2302.03024* (2023).
- [201] YANG, W., XIE, Y., LIN, A., LI, X., TAN, L., XIONG, K., LI, M., AND LIN, J. End-to-end open-domain question answering with bertserini. *arXiv preprint arXiv:1902.01718* (2019).
- [202] YE, H., ZHANG, J., LIU, S., HAN, X., AND YANG, W. Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models. *arXiv preprint arXiv:2308.06721* (2023).
- [203] YOO, S., KIM, E., JUNG, D., LEE, J., AND YOON, S. Improving visual prompt tuning for self-supervised vision transformers. *arXiv preprint arXiv:2306.05067* (2023).
- [204] YOU, Q., JIN, H., WANG, Z., FANG, C., AND LUO, J. Image captioning with semantic attention. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (2016), pp. 4651–4659.
- [205] YU, Q., HE, J., DENG, X., SHEN, X., AND CHEN, L.-C. Convolutions die hard: Open-vocabulary segmentation with single frozen convolutional clip. *arXiv preprint arXiv:2308.02487* (2023).
- [206] ZADOURI, T., ÜSTÜN, A., AHMADIAN, A., ERMIŞ, B., LOCATELLI, A., AND HOOKER, S. Pushing mixture of experts to the limit: Extremely parameter efficient moe for instruction tuning. *arXiv preprint arXiv:2309.05444* (2023).
- [207] ZAKEN, E. B., RAVFOGEL, S., AND GOLDBERG, Y. Bitfit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. *arXiv preprint arXiv:2106.10199* (2021).
- [208] ZELLERS, R., HOLTZMAN, A., BISK, Y., FARHADI, A., AND CHOI, Y. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (2019).
- [209] ZENG, B., LI, S., FENG, Y., LI, H., GAO, S., LIU, J., LI, H., TANG, X., LIU, J., AND ZHANG, B. Ipdreamer: Appearance-controllable 3d object generation with image prompts. *arXiv preprint arXiv:2310.05375* (2023).
- [210] ZENG, G., ZHANG, P., AND LU, W. One network, many masks: Towards more parameter-efficient transfer learning. *arXiv preprint arXiv:2305.17682* (2023).
- [211] ZHANG, B., AND SENNRICH, R. Root mean square layer normalization. *Advances in Neural Information Processing Systems* 32 (2019).
- [212] ZHANG, H., LIU, X., AND ZHANG, J. Summit: Iterative text summarization via chatgpt. *arXiv preprint arXiv:2305.14835* (2023).
- [213] ZHANG, J. O., SAX, A., ZAMIR, A., GUIBAS, L., AND MALIK, J. Side-tuning: a baseline for network adaptation via additive side networks. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part III* 16 (2020), Springer, pp. 698–714.
- [214] ZHANG, L., RAO, A., AND AGRAWALA, M. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 3836–3847.
- [215] ZHANG, L., ZHANG, L., SHI, S., CHU, X., AND LI, B. Lora-fa: Memory-efficient low-rank adaptation for large language models fine-tuning. *arXiv preprint arXiv:2308.03303* (2023).
- [216] ZHANG, M., SHEN, C., YANG, Z., OU, L., YU, X., ZHUANG, B., ET AL. Pruning meets low-rank parameter-efficient fine-tuning. *arXiv preprint arXiv:2305.18403* (2023).
- [217] ZHANG, Q., CHEN, M., BUKHARIN, A., HE, P., CHENG, Y., CHEN, W., AND ZHAO, T. Adaptive budget allocation for parameter-efficient fine-tuning. *arXiv preprint arXiv:2303.10512* (2023).
- [218] ZHANG, R., FANG, R., ZHANG, W., GAO, P., LI, K., DAI, J., QIAO, Y., AND LI, H. Tip-adapter: Training-free clip-adapter for better vision-language modeling. *arXiv preprint arXiv:2111.03930* (2021).
- [219] ZHANG, R., GUO, Z., ZHANG, W., LI, K., MIAO, X., CUI, B., QIAO, Y., GAO, P., AND LI, H. Pointclip: Point cloud understanding by clip. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022), pp. 8552–8562.
- [220] ZHANG, R., HAN, J., ZHOU, A., HU, X., YAN, S., LU, P., LI, H., GAO, P., AND QIAO, Y. Llama-adapter: Efficient fine-tuning of language models with zero-init attention. *arXiv preprint arXiv:2303.16199* (2023).
- [221] ZHANG, R., QIANG, R., SOMAYAJULA, S. A., AND XIE, P. Autolora: Automatically tuning matrix ranks in low-rank adaptation based on meta learning. *arXiv preprint arXiv:2403.09113* (2024).
- [222] ZHANG, Y., ZHOU, K., AND LIU, Z. Neural prompt search, 2022.
- [223] ZHANG, Z., SHENG, Y., ZHOU, T., CHEN, T., ZHENG, L., CAI, R., SONG, Z., TIAN, Y., RÉ, C., BARRETT, C., ET AL. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *Advances in Neural Information Processing Systems* 36 (2024).
- [224] ZHANG, Z.-R., TAN, C., XU, H., WANG, C., HUANG, J., AND HUANG, S. Towards adaptive prefix tuning for parameter-efficient language model fine-tuning. *arXiv preprint arXiv:2305.15212* (2023).
- [225] ZHAO, B., TU, H., WEI, C., MEI, J., AND XIE, C. Tuning layernorm in attention: Towards efficient multi-modal llm finetuning. *arXiv preprint arXiv:2312.11420* (2023).
- [226] ZHAO, H., FU, J., AND HE, Z. Prototype-based hyperadapter for sample-efficient multi-task tuning. *arXiv preprint arXiv:2310.11670* (2023).
- [227] ZHAO, H., HE, R., XIAO, M., AND XU, J. Infusing hierarchical guidance into prompt tuning: A parameter-efficient framework for multi-level implicit discourse relation recognition. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023), pp. 6477–6492.
- [228] ZHAO, J., ZHANG, Z., CHEN, B., WANG, Z., ANANDKUMAR, A., AND TIAN, Y. Galore: Memory-efficient llm training by gradient low-rank projection. *arXiv preprint arXiv:2403.03507* (2024).
- [229] ZHAO, Z., HU, L., ZHAO, H., SHAO, Y., AND WANG, Y. Knowledgeable parameter efficient tuning network for commonsense question answering. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023), pp. 9051–9063.
- [230] ZHOU, B., ZHAO, H., PUIG, X., FIDLER, S., BARRIUSO, A., AND TORRALBA, A. Scene parsing through ade20k dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (2017), pp. 633–641.
- [231] ZHOU, H., WAN, X., VULIĆ, I., AND KORHONEN, A. Autopeft: Automatic configuration search for parameter-efficient fine-tuning. *arXiv preprint arXiv:2301.12132* (2023).
- [232] ZHOU, K., YANG, J., LOY, C. C., AND LIU, Z. Conditional prompt learning for vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022), pp. 16816–16825.
- [233] ZHOU, K., YANG, J., LOY, C. C., AND LIU, Z. Learning to prompt for vision-language models. *International Journal of Computer Vision* 130, 9 (2022), 2337–2348.
- [234] ZHOU, T., AND TAO, D. Godec: Randomized low-rank & sparse matrix decomposition in noisy case. In *Proceedings of the 28th International Conference on Machine Learning, ICML 2011* (2011).
- [235] ZHOU, Z., WEI, X., ZHANG, J., AND SUN, G. {PetS}: A unified framework for {Parameter-Efficient} transformers serving. In *2022 USENIX Annual Technical Conference (USENIX ATC 22)* (2022), pp. 489–504.
- [236] ZHU, B., NIU, Y., HAN, Y., WU, Y., AND ZHANG, H. Prompt-aligned gradient for prompt tuning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 15659–15669.
- [237] ZHU, D., CHEN, J., SHEN, X., LI, X., AND ELHOSEINY, M. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592* (2023).

- [238] ZHU, Q., LI, B., MI, F., ZHU, X., AND HUANG, M. Continual prompt tuning for dialog state tracking. *arXiv preprint arXiv:2203.06654* (2022).
- [239] ZHU, W., LIU, H., DONG, Q., XU, J., KONG, L., CHEN, J., LI, L., AND HUANG, S. Multilingual machine translation with large language models: Empirical results and analysis. *arXiv preprint arXiv:2304.04675* (2023).
- [240] ZHU, W., AND TAN, M. Spt: Learning to selectively insert prompts for better prompt tuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (2023), pp. 11862–11878.
- [241] ZHU, X., ZHANG, R., HE, B., GUO, Z., ZENG, Z., QIN, Z., ZHANG, S., AND GAO, P. Pointclip v2: Prompting clip and gpt for powerful 3d open-world learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2023), pp. 2639–2650.
- [242] ZHU, Y., FENG, J., ZHAO, C., WANG, M., AND LI, L. Counter-interference adapter for multilingual machine translation. *arXiv preprint arXiv:2104.08154* (2021).
- [243] ZHUANG, Y., YU, Y., WANG, K., SUN, H., AND ZHANG, C. Toolqa: A dataset for llm question answering with external tools. *arXiv preprint arXiv:2306.13304* (2023).